

SE - Overview of IT Industry

Session 5: Git & Version Control Why version control is needed. Git concepts: Repository, Commit, Branch, Merge, Push, Pull. GitHub overview: Forks, Pull Requests, Collaboration. Trainer Demo: Initialize a local Git repo (git init, git add ., git commit) Push to GitHub. AI Integration: Use GitHub Copilot Chat to explain git errors in real-time. Depth: Students should confidently push their first small project to GitHub. Task :

Question :- 1

Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Ans :-

Initialize a repository :- git init

Stage all files :- git add

Commit with a message :- git commit -m "First commit"

Push to remote repository :- git push origin main

Pull latest changes :- git pull origin main

Question :- 2

What is the difference between git fetch and git pull? Answer in 2 lines.

Ans :-

Git fetch :- git fetch downloads the latest changes from the remote repository but does not merge them.

Git pull :- git pull downloads the changes and automatically merges them into your current branch.

Question :- 3

Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans :-

<https://github.com/harshdav0054/TOPS-Demo>

Question :- 4

What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Ans :-

Fork:- Creates your own copy of someone else's GitHub repository.

Pull Request (PR):- A request to merge your changes into the original repository.

Question :- 5

Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans :-

Step 1:- Create and switch to a new branch named feature-test.

Command:

```
git checkout -b feature-test
```

Step 2:- Make changes to the project files and save them.

Step 3:- Add the modified files to the staging area.

Command:

```
git add .
```

Step 4:- Commit the changes with a message.

Command:

```
git commit -m "Added test file"
```

Step 5:- Switch back to the main branch.

Command:

```
git checkout main
```

Step 6:- Merge the feature-test branch into the main branch.

Command:

```
git merge feature-test
```

Result:- The changes made in the feature-test branch were successfully merged into the main branch.